Removable collisions in the dS_3 crossing kernel: merged series and a controlled heavy-mass factor

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Abstract

We reorganize the known LPRS scalar crossing formula to remove a numerically unstable pair of Gamma poles. A termwise merged series has a finite digamma-weighted limit on each real collision surface. In a specified transfer window we prove an explicit absolute remainder bound for one heavy-mass Omega factor, and a direct-sum tail bound that avoids an uncertified asymptotic acceleration. A pointwise logarithmic near-collision scaling follows for this factor. These results do not establish the distributional limit of the full Racah transform or the Bose free-observer limit. Numerical failures, including a prototype precision bug, remain in the record.

1 Physical scope and the removable surfaces

The target remains the local $G_N = 0$, radius-one model with proper $SO^+(3,1)$, BD scalar observers and KG-normalized matter. Observer labels exceed a fixed $s > 0$; the two-creation sector uses the previously defined half-period regulator and physical mass $\sqrt{1+\nu^2}$. Here we study the scalar matrix element of the full crossing transform, not a separately unitary scalar transform. No state, operator, or physical spectral cutoff is changed.

In the LPRS formula [1], use $h_i = (1+\mathrm{i}\nu_i)/2$, $h = (1+\mathrm{i}b)/2$, $h' = (1+\mathrm{i}d)/2$ and external labels (a,q,c,p). Both chiral weights are equal for the scalar principal series; they are not complex conjugates. Each Omega factor in (3.38) is a sum

$$\Omega(h,h';h_i) = \Gamma(z)A(z) + \Gamma(-z)B(z), \qquad z = h - h' + h_1 + h_3 - 1.$$

The full symbol contains h and $1-h$, and h' and $1-h'$. For positive real labels the possible $z = 0$ surfaces are

$$a + c = b + d, \qquad d = a + c + b, \qquad b = a + c + d.$$

These are removable singularities of the two-term representation. They are not physical poles at finite real mass. The original Euler product integral (3.37) is locally integrable: its absolute endpoint powers are $x^{-1/2}(1-x)^{-1/2}$, multiplied at worst by logarithmic hypergeometric endpoint factors. On compact parameter sets it gives continuity through these surfaces.

2 Exact merger of the two series

Fix H and h_i, put $h'_0 = H + h_1 + h_3 - 1$ and $h' = h'_0 - z$. Define four upper parameters

$$(A_1, A_2, A_3, A_4) = (H + h_1 + h_2 - 1,\ H + h_3 + h_4 - 1,\ H + h_3 - h_4,\ H + h_1 - h_2).$$

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All have real part $1/2$ on our real-label slice. Let

$$C(z) = \frac{\Gamma(1+z)\Gamma(1-z)\Gamma(2h_0'-2z)\Gamma(2H)}{\Gamma(A_1-z)\Gamma(A_2-z)\Gamma(A_3)\Gamma(A_4)}.$$

Direct cancellation of Gamma factors in the two known ${}_4F_3$ series yields

$$\Omega = C(z)\sum_{n=0}^{\infty}\frac{G_n^A(z)-G_n^B(z)}{z}, \tag{1}$$

$$G_n^A(z) = \frac{\prod_{j=1}^4\Gamma(n+A_j-z)}{\Gamma(n+2h_0'-2z)\Gamma(n+2H-z)\Gamma(n+1-z)\Gamma(n+1)}, \tag{2}$$

$$G_n^B(z) = \frac{\prod_{j=1}^4\Gamma(n+A_j)}{\Gamma(n+2h_0'-z)\Gamma(n+2H)\Gamma(n+1+z)\Gamma(n+1)}. \tag{3}$$

The individual G's have the same leading n^{-2} behavior. Their difference divided by z is $2n^{-3}+O(n^{-4})$ for fixed parameters, with the $z=0$ value interpreted continuously. Thus the merged series converges absolutely and locally uniformly near a real collision. This is an algebraic reorganization of an existing formula, not a new definition of the physical kernel.

At $z=0$, the two regular prefactors $A(0)$ and $B(0)$ coincide. Expanding $\Gamma(\pm z)$ gives $A'(0)-B'(0)-2\gamma A(0)$. Equivalently, termwise differentiation of (1) gives

$$\Omega_0 = \sum_{n=0}^{\infty} t_n\, q_n, \tag{4}$$

$$t_n = \frac{\prod_{j=1}^4(A_j)_n}{(2h_0')_n(2H)_n(n!)^2}, \tag{5}$$

$$q_n = \psi(n+2h_0')+\psi(n+2H)+2\psi(n+1)-\sum_{j=1}^{4}\psi(n+A_j). \tag{6}$$

The digamma bracket starts as $2/n+O(n^{-2})$. The local uniform convergence justifies the parameter derivative. No mass displacement is used to define the exact-surface value.

For computation away from zero, use $G_n^B\,\mathrm{expm1}(\log G_n^A-\log G_n^B)/z$. Logarithmic Gamma differences and successive ratios must retain their branches continuously. Arbitrary precision still needs guard digits when $|z|$ is tiny. Gamma-ratio/Bernoulli and Hurwitz-zeta acceleration is available [2], but its truncated large-parameter remainder is not automatically controlled.

3 A proved heavy-mass window

Now select the colliding factor $H=(1-\mathrm{i}b)/2$ with

$$c=M,\quad b=M+u,\quad a=M+u+v,\quad d=M+v,\qquad |u|+p\le 1,\quad |v|+q\le 1.$$

Assume all physical observer labels remain above s. Then $h_0'=(1+\mathrm{i}d)/2$ and

$$(A_1,A_2,A_3,A_4)=\left(\frac{1+\mathrm{i}(v+q)}{2},\frac{1+\mathrm{i}(p-u)}{2},\frac{1-\mathrm{i}(u+p)}{2},\frac{1+\mathrm{i}(v-q)}{2}\right).$$

Write $A_j=1/2+\mathrm{i}\alpha_j$; the window ensures $|\alpha_j|\le 1/2$. For $m=\min(b,d)>0$, the coefficient ratio obeys

$$\left|\frac{t_{n+1}}{t_n}\right|\le\left(\frac{n+1}{n+2}\right)^2\frac{(n+1)^2}{(n+1)^2+m^2}.$$

To see this, bound each $|n + A_j|^2$ by $n^2 + n + 1/2$, and use $(x^2 - x + 1/2)(x + 1) \le x^3$ for $x = n + 1 \ge 1$. Consequently

$$|t_n| \le \frac{1}{(n+1)^2} \prod_{j=1}^{n} \frac{j^2}{j^2 + m^2} \le \frac{1}{(n+1)^2(1+m^2)}, \qquad n \ge 1. \tag{7}$$

The digamma difference series gives the uniform bound

$$|q_n| \le B_M, \qquad B_M = 2 + \operatorname{arsinh} b + \operatorname{arsinh} d + 4 \log 2 \sum_j |A_j - 1|.$$

For the upper parameters, use $\sum_{k\ge 0}[(k + 1/2)(k + 1)]^{-1} = 4 \log 2$. For a denominator parameter $1+\mathrm{i}y$, the difference from $\psi(n+1)$ is bounded by $\sum_{j\ge 1} |y|/[j\sqrt{j^2 + y^2}] \le 1+\operatorname{arsinh} |y|$. Combining these estimates proves

$$\Omega_0 = q_0 + R_M, \qquad |R_M| \le \frac{(\zeta(2) - 1)B_M}{1 + m^2}, \tag{8}$$

$$q_0 = \psi(1 - \mathrm{i}b) + \psi(1 + \mathrm{i}d) + 2\psi(1) - \sum_j \psi(A_j). \tag{9}$$

This is an explicit $O(\log M/M^2)$ absolute remainder for fixed transfers in the stated window. Standard digamma asymptotics then imply $\Omega_0 = 2 \log M + 2\psi(1) - \sum_j \psi(A_j) + O(M^{-1})$.

More usefully, summing (4) only through $n = N - 1$ has the analytic exact-arithmetic tail bound

$$\left| \Omega_0 - \sum_{n=0}^{N-1} t_n q_n \right| \le \frac{B_M}{N} \prod_{j=1}^{N} \frac{j^2}{j^2 + m^2}. \tag{10}$$

Here $\sum_{n=N}^{\infty}(n + 1)^{-2} \le 1/N$. Unlike a fitted zeta tail, (10) is a proved bound on the omitted series in this window. It does not bound floating-point error.

4 Neighbouring masses and logarithmic scaling

Let $d = M + v + r$ so that $z = -\mathrm{i}r/2$. Suppose both $|\operatorname{Im} A_j| \le 1/2$ and $|\operatorname{Im}(A_j - z)| \le 1/2$. The two unmerged hypergeometric series are ${}_4F_3$ with numerator sets $A_j - z$ and A_j, and denominator sets

$$(2h', 2H - z, 1 - z), \qquad (2h'_0 - z, 2H, 1 + z).$$

Define

$$\begin{aligned} a_0(z) &= \frac{\Gamma(2H)\Gamma(A_3 - z)\Gamma(A_4 - z)}{\Gamma(A_3)\Gamma(A_4)\Gamma(2H - z)}, \\ b_0(z) &= \frac{\Gamma(2h')\Gamma(A_1)\Gamma(A_2)}{\Gamma(A_1 - z)\Gamma(A_2 - z)\Gamma(2h'_0 - z)}, \\ m_A &= \min(|M + v + r|, |M + u - r/2|), \\ m_B &= \min(|M + v + r/2|, |M + u|). \end{aligned}$$

For $r \ne 0$ and $m_A, m_B > 0$, the same ratio proof gives

$$|\Omega - \Gamma(z)a_0 - \Gamma(-z)b_0| \le (\zeta(2) - 1) \left(\frac{|\Gamma(z)a_0|}{1 + m_A^2} + \frac{|\Gamma(-z)b_0|}{1 + m_B^2} \right). \tag{11}$$

Each hypergeometric tail has bound $N^{-1} \prod_{j=1}^{N} j^2/(j^2 + m_{A,B}^2)$, multiplied by its corresponding Gamma prefactor. These bounds grow as $1/|r|$ near zero; use (10) at exact collision.

For fixed $\lambda \neq 0$, set $r = \lambda/\log M$. Use the strict interior of the transfer window so the shifted constraints eventually hold. Gamma-ratio asymptotics in the sectors of $1 \pm \mathrm{i}M$ give

$$a_0(z) = (-\mathrm{i}M)^z[1 + O(z) + O(M^{-1})], \qquad b_0(z) = (\mathrm{i}M)^{-z}[1 + O(z) + O(M^{-1})].$$

Since $\Gamma(\pm z) = \pm z^{-1} + O(1)$, the leading pair divided by $2\log M$ tends to $\sin(\lambda/2)/(\lambda/2)$. The error (11) is $O(\log M/M^2)$. Together with (8) at $\lambda = 0$ this proves the pointwise statement

$$\frac{\Omega(r = \lambda/\log M)}{2\log M} \longrightarrow \frac{\sin(\lambda/2)}{\lambda/2}, \qquad \lambda \text{ fixed.} \tag{12}$$

The right side is one at zero. We do not claim a uniform error in λ, distributional convergence of the full $6j$, or a delta replacement inside a physical mass integral.

5 Numerical evidence and retained failures

For $p = q = 0.6$, $u = 0.3$, $v = -0.2$, the complete physical kernel on $a + c = b + d$ has the following pointwise reference values:

M	2	4	8	16
U_{00}	0.96379723	1.15801422	1.36981862	1.58731944

A separate $M = 32$ reference gives 1.80680785. These are point values, not a mass-integrated Bose observable. The initial study also retains offsets $r = 0.1, 0.4$ at $M = 2, 4, 8, 16$. No full-kernel limiting profile is inferred from these samples.

For the complete $M = 32$ kernel, merely increasing arithmetic precision does not repair an inadequate asymptotic tail. At 110 decimal digits, the merged evaluator with $(N, K) = (512, 24)$ has relative error about $2.42\,10^{24}$ against the separate LPRS reference. The successive settings $(1024, 32)$, $(2048, 40)$, $(4096, 48)$ and $(8192, 56)$ give approximately $5.78\,10^{6}$, $1.74\,10^{-13}$, $5.39\,10^{-19}$ and $8.60\,10^{-25}$. All attempts are retained. These are sample-specific empirical errors, not a uniform accuracy claim for the algorithm.

For the single controlled Omega factor, the rigorous direct-sum tail at $M = 32$, $r = 0$, $N = 128$ is about $2.62\,10^{-39}$ in exact arithmetic. At much larger M this tail becomes far smaller than the available rounding precision; it must not be reported as the total numerical error. Computations at 70 and 100 decimal digits are stored separately for the same series cutoff.

Convergence in (12) is slow: at $M = 1024$ the complex scaled value for $\lambda = 0$ is approximately $1.31857 + 0.08411i$, whose absolute difference from one is 0.32949. For $\lambda = 1$ it is approximately $1.27753 + 0.09388i$, with absolute difference 0.33222 from $\sin(1/2)/(1/2)$. The existence of the asymptotic limit does not imply these masses are already close to it.

The first collision prototype accidentally used Python binary64 division in one Bernoulli coefficient before multiplication by an arbitrary-precision value. At a low-mass sample it produced an Omega bias about $2.82\,10^{-27}$ which was shared by precision refinements. The old source and outputs remain unchanged. The corrected merged module uses arbitrary-precision coefficients; an independent parameter derivative reduces that sample discrepancy to about $9.74\,10^{-58}$.

All three exact surfaces have been checked against that parameter derivative, including matter-flip Hermiticity. Near-surface offsets 10^{-8} and 10^{-25} and an independent Euler integral were also checked. The nine diagnostics pass their stated tolerances; none is an interval proof of the arbitrary-precision library. An additional 100-digit comparison of the direct-sum analytic bounds with complete hypergeometric or parameter-derivative evaluations is still running at this checkpoint. It is not counted as a passed test. No old physical wave-packet grid is overwritten.

6 What remains before the Bose free limit

The Bose target remains $N_{\mathrm{free}}(t) = L_p(0)^2 + |L_p(t)|^2$ and $D_{\mathrm{free}}(t) = 0$ under an explicitly specified heavy-observer preparation protocol. The present analytic bound covers only one of four Omega factors in the complete LPRS expression. Other factors and shadow terms can exponentially amplify individual absolute errors and cancel among themselves. Their phases and physical normalization must be controlled jointly. Pointwise kernel values cannot justify exchanging a mass integral with a heavy-mass limit.

The next step is a uniform analysis of the full prefactored crossing kernel against actual creation amplitudes, with mass tails and cancellation-aware absolute bounds. A different integral representation is preferable if the factorized representation cannot supply such control. Complete Fock domains, finite gravity, and a microscopic de Sitter holographic dual remain open.

References

[1] J. Liu, E. Perlmutter, V. Rosenhaus and D. Simmons-Duffin, *d-dimensional SYK, AdS Loops, and* $6j$ *Symbols*, arXiv:1808.00612, v3, Eqs. (3.35)–(3.38).

[2] NIST Digital Library of Mathematical Functions, Sec. 5.11 (Gamma-ratio and digamma expansions), consulted 31 October 2026.